

Business Setup For LED LIGHTING ASSEMBLY

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Light-emitting diodes (LEDs) are semiconductor devices that emit visible light when electric current passes through them. Compared to conventional lighting systems, these are smaller, have a longer operating life and involve a lower cost of ownership. Available in a wide range of colours, LED light bulbs are more durable and offer comparable or better light quality than other types of lighting.

Residential LED lights, especially Energy Star rated products, consume at least 75 per cent less energy and last 25 times longer than incandescent lights. These also use significantly less power—a typical 84-watt fluorescent light can be replaced with a 36-watt LED to give the same level of light output.

The LED lamp market is projected to grow at 25 per cent annually to reach \$25 billion in 2023. Hence LED light manufacturing is a profitable investment opportunity for entrepreneurs.

Full-scale LED light manufacturing requires setting up of a large-scale factory with a complex production process. However, it is possible for small-scale entrepreneurs to set up LED light assembly units as a small- or medium-scale manufacturing operation.

Process

LED-based lighting system cum LED lamp assembly consists of the following steps:

1. Procure/import milliwatt-rated LED chips, circuit and other mounting devices
2. Embed milliwatt-rated LED chips on the PCB board with the rectifier circuit, filter circuit, etc
3. Fit the PCB board with a holder cap and plastic modules fitted with Smokey reflector to form a compact unit
4. Test the assembled LED lighting system and package

Raw materials

For assembly of LED-based lighting systems up to 10W you may require:

1. LED chips
2. Rectifier circuit with filter
3. Heat-sink devices
4. Metallic cap holder
5. Plastic body
6. Reflector plastic glass
7. Connecting wire
8. Soldering flux
9. Miscellaneous items
10. Packaging material

Equipment required

LED light manufacturing or assembly is a complex process. Machines need to be selected on the basis of the specific LED type that is being produced and the raw material being used. However, major machines include:

1. LED PCB assembly machine
2. LED lights assembly machine
3. High-speed LED mounting machine

TABLE I
TOTAL FIXED CAPITAL

1. MACHINERY AND EQUIPMENT COST

Description	Quantity	Value (₹)
Component forming machine	2	100,000
Soldering machine	10	5000
Digital multimeter	2	8000
Continuity tester	10	1000
Sealing machine	1	10,000
Packaging machine	1	15,000
LCR meter	2	20,000
Small drilling machine	1	10,000
Lux meter	1	40,000
Oscilloscope	1	60,000
Personnel computer with UPS and printer	1	80,000
Miscellaneous items	—	10,000
Total	—	359,000

2. OTHER FIXED ASSETS

Description	Value (₹)
Electrification charges @ 10 per cent of machinery cost	35,900
Office equipment, furniture and working tables, etc	20,000
Tools, jigs and fixtures	20,000
Pre-operative expenses	5000
Miscellaneous items	5000
Total	85,900
Total fixed capital (1+2)	444,900